

3.4 P

Pr 91

$$R = 9 \text{ cm} = 0.09 \text{ m}$$

$$(1) \quad \omega_1 = 2000 \text{ r/min} = \frac{100}{3} \text{ r/s}, \quad \omega_1 = 2\pi n_1 = \frac{200\pi}{3} \text{ rad/s},$$

$$~~v = \omega R~~ \quad v = \omega_1 R = 6\pi \text{ m/s}$$

$$\textcircled{2} \quad a_n = \omega_1^2 R = 400\pi^2 \text{ m/s}^2$$

$$(2) \quad \omega_2 = 2\pi n_2 = \frac{400\pi}{3} \text{ rad/s}, \quad \alpha = \frac{\omega_2 - \omega_1}{\Delta t} = \frac{2\pi}{9} \text{ rad/s}^2$$

$$a_t = \alpha R = \frac{\pi}{50} \text{ m/s}^2$$

$$a_n' = \omega_2^2 R = 1600\pi^2 \text{ m/s}^2, \quad a = \sqrt{a_t^2 + (a_n')^2} = 1.58 \times 10^4 \text{ m/s}^2$$

5.6 1) $n_A = 600 \text{ r/min} = 10 \text{ r/s}$, $\alpha = \frac{4\pi}{5} \text{ rad/s}^2$, ~~$\Delta t =$~~

$n_B = \frac{r_1}{r_2} n_A = 4 \text{ r/s}$, $\omega_A = 20\pi \text{ rad/s}$, $\omega_B = \frac{r_1}{r_2} \omega_A = 8\pi \text{ rad/s}$.

$$\Delta t = \frac{\omega_B}{\alpha} = 10 \text{ s}$$

2) $\omega_A' = \frac{300 \times 2\pi}{60} = 10\pi \text{ rad/s}$. $\beta = \frac{\omega_A' - \omega_A}{\Delta t} = -\frac{\pi}{6} \text{ rad/s}^2$, 減速.

$$\Delta \theta = \frac{\omega_A + \omega_A'}{2} \Delta t = 900\pi \text{ rad}, N = \frac{\Delta \theta}{2\pi} = 450 \text{ 转}$$

5.10 设 $OB = x$, 由余弦定理: $l^2 = r^2 + x^2 - 2rx \cos \theta$.

$$\Rightarrow x^2 - (2r \cos \theta) \cdot x + r^2 - l^2 = 0, \Delta = 4l^2 - 4r^2 \sin^2 \theta.$$

解得 $x = r \cos \theta + \sqrt{l^2 - r^2 \sin^2 \theta}$ (~~显然~~ $x > r \geq r \cos \theta - \sqrt{l^2 - r^2 \sin^2 \theta}$.)

两边对 t 求导 $\Rightarrow \frac{dx}{dt} = v = r \frac{d \cos \theta}{d \theta} \frac{d \theta}{dt} + \frac{-r^2}{2\sqrt{l^2 - r^2 \sin^2 \theta}} \cdot \frac{d \sin^2 \theta}{d \sin \theta} \frac{d \sin \theta}{d \theta} \frac{d \theta}{dt}$.

由 $\omega = \frac{d \theta}{dt}$ 代入: $= -r \omega \sin \theta - \frac{r^2 \omega \sin 2 \theta}{2\sqrt{l^2 - r^2 \sin^2 \theta}}$.

负号为 I、II 象限 θ 增加时, 活塞向 x 轴负向运动

6.5. 鼓轮、绳质量不计, 转动惯量为 $J=0$.

设角加速度 α . 张力为 T_1, T_2 .

假设鼓轮向顺时针转动, 有
$$\begin{cases} T_1 - m_1 g = m_1 a_1 & (1) \\ m_2 g - T_2 = m_2 a_2 & (2) \end{cases}$$

又
$$\begin{cases} a_1 = \alpha r_1 & (3) \\ a_2 = \alpha r_2 & (4) \end{cases}$$
 且由力矩平衡: $T_2 r_2 - T_1 r_1 = J \alpha$.
$$\Rightarrow \underline{T_1 r_1 = T_2 r_2} \quad (5)$$

①②③④⑤ 联立得
$$\alpha = \frac{(m_2 r_2 - m_1 r_1)}{m_1 r_1^2 + m_2 r_2^2} g.$$

6.6. 考虑转动惯量分别为 J_1, J_2 . 总转动惯量 $J = J_1 + J_2$.

$$\begin{cases} T_1 - m_1 g = m_1 \alpha r_1 \\ m_2 g - T_2 = m_2 \alpha r_2 \\ T_2 r_2 - T_1 r_1 = (J_1 + J_2) \alpha. \end{cases}$$

解得
$$\alpha = \frac{(m_2 r_2 - m_1 r_1)}{J_1 + J_2 + m_1 r_1^2 + m_2 r_2^2} g. \quad (6)$$

$$\begin{cases} a_1 = \alpha r_1 \\ a_2 = \alpha r_2 \end{cases} \quad \begin{cases} T_1 = m_1 (g + \alpha r_1) \\ T_2 = m_2 (g - \alpha r_2) \end{cases} \quad \Leftarrow \text{将 (6) 代入即得}$$

6.17 转轴过中心, 角动量守恒

$$m = 0.25 \text{ kg}, M = 1 \text{ kg}, L = 1 \text{ m},$$

初始球在C点, $r_1 = 0$, $\omega_1 = 10 \text{ rad/s}$; 末状态 $r_2 = \frac{1}{2}L = 0.5 \text{ m}$.

$$J_{\text{细管}} = \frac{1}{12}ML^2 = \frac{1}{12} \text{ kg} \cdot \text{m}^2$$

$$J_1 = J_{\text{细管}} + m \cdot r_1^2 = \frac{1}{12} \text{ kg} \cdot \text{m}^2$$

$$J_2 = J_{\text{细管}} + m r_2^2 = \frac{7}{48} \text{ kg} \cdot \text{m}^2$$

角动量守恒, $L_1 = L_2$;

$$J_1 \omega_1 = J_2 \omega_2 \Rightarrow \omega_2 = \frac{40}{7} \text{ rad/s}.$$

b.19 $L = 1\text{m}, m = 2.5\text{kg}.$

(1) $L = F \cdot L \Delta t = 2\text{kg} \cdot \text{m}^2/\text{s}$

(2) $J = \frac{1}{3}mL^2 = \frac{5}{6}\text{kg} \cdot \text{m}^2$, 由 $L = J\omega \Rightarrow \omega = \frac{L}{J} = \frac{12}{5}\text{rad/s}.$

由机械能守恒, $\frac{1}{2}J\omega^2 = mgh_c$

得 $h_c = 0.098\text{m}$

$h = 2h_c = 0.196\text{m}.$

b.21 $m_A = m, m_B = 4m, r_A = r, r_B = 2r, \omega_A = \omega, \omega_B = 0$
 $(1) J_A = \frac{1}{2}mr^2$
 $J_B = \frac{1}{2}(4m)(2r)^2 = 8mr^2$

(2) $L_1 = J_A\omega_1 + J_B \cdot 0 = \frac{1}{2}m\omega_1 r^2$

$J' = J_A + J_B = \frac{17}{2}mr^2$

由 $L_1 = L_2 \Rightarrow \frac{1}{2}m\omega_1 r^2 = \frac{17}{2}m\omega' r^2$

$\Rightarrow \omega' = \frac{1}{17}\omega_1$

(3) $E_{k1} = \frac{1}{2}J_A\omega_1^2 = \frac{1}{4}mr^2\omega_1^2$

$E_{k2} = \frac{1}{2}J'(\omega')^2 = \frac{1}{88}mr^2\omega_1^2$

$\Delta E_k = \frac{4}{17}mr^2\omega_1^2$

$$9. \Delta x = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

$$(1) J = \frac{1}{12} ml^2 + m \cdot \left(\frac{l}{6}\right)^2 = \frac{1}{9} ml^2$$

$$(2) M = mg \Delta x = \frac{1}{6} mgl$$

$$\text{由 } M = J \alpha_1 \Rightarrow \alpha_1 = \frac{3g}{2l}$$

$$(2) mgh = \frac{1}{2} J \omega^2 \Rightarrow \omega = \sqrt{\frac{3g}{l}}$$

垂直位置, 重力通过转轴, $M = 0$

$$\Rightarrow \alpha_2 = 0$$

(3) 完全非弹性碰撞满足角动量守恒;

$$J' = J + m \cdot \left(\frac{2l}{3}\right)^2 = \frac{5}{9} ml^2$$

$$\text{由 } J \omega = J' \omega' \Rightarrow \omega' = \frac{1}{5} \omega = \frac{1}{5} \sqrt{\frac{3g}{l}}$$

$$\Delta h_A = \frac{l}{6} (1 - \cos \theta), \Delta h_C = \frac{2l}{3} (1 - \cos \theta)$$

$$\Rightarrow \text{机械能守恒, } \frac{1}{2} J' (\omega')^2 = mg (\Delta h_A + \Delta h_C) \text{ 解得 } \theta = \arccos\left(\frac{24}{25}\right)$$

$$10. f(r) = -kr \Rightarrow \underline{dF = -kr dr}$$

微元法: 取半径为 r 宽度 dr 的环形面积:

$$dS = 2\pi r dr$$

$$\text{扭矩 } dM_f = r \cdot dF = -kr^2 dS = -2k\pi r^3 dr$$

$$\Rightarrow \int dM_f = M_f = -2k\pi \int_0^R r^3 dr = -\frac{k\pi}{2} R^4$$

又由 $M_f = J \alpha$, $J = \frac{1}{2} m R^2$ 知

$$-\frac{k\pi}{2} R^4 = \frac{1}{2} m R^2 \frac{d\omega}{dt}$$

$$\Rightarrow \frac{d\omega}{dt} = -\frac{\pi k R^2}{m}$$

$$\Rightarrow t = \frac{\omega_0}{|\alpha|} = \frac{m \omega_0}{\pi k R^2}$$

$$(2) \theta = \frac{\omega_0^2}{2\alpha} = \frac{m \omega_0^2}{2\pi k R^2}$$